

COMMENTARY



Beyond prediction: comments on the format of natural intelligence

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Parr et al. (2025) treat syntax as rules over temporal sequences – a non-Markovian process that can be captured by autoregressive or hierarchical-temporal models. This move quietly collapses into two distinct representational layers:

Syntax: Hierarchical constituent geometry formed in an abstract workspace. This is *never linear*.

Externalized phonology: How syntax is transformed from a hierarchical tree-structure to a string. This is *always linear*.

By equating syntax with non-Markovian statistics over strings, the authors risk inheriting two problems:

- (1) **Linguistic adequacy:** Many phenomena (binding, displacement, island effects, structural ambiguity, *wh*-movement, topicalization, recursion limits) are defined over trees, not strings (Adger, 2003; Everaert et al., 2015; Marcolli & Larson, 2025; Murphy, 2020, 2025; Murphy & Leivada, 2022; Reuland, 2011), and are not computed over sequential rules, but over *structural/hierarchical distance* (Figure 1).
- (2) **Neurobiological adequacy:** In addition to tracking surface statistics, the brain *also* tracks constituent boundaries and tree-geometric depth (McCarty et al., 2023; Murphy, 2024; Weissbart & Martin, 2024; Woolnough et al., 2023).

This human bias to impose tree-structures may be innate (Crain et al., 2017; Perkins & Lidz, 2021; Yang et al., 2017). I will assume here (not without contention) that trees are a privileged representational format (see Dehaene et al., 2022; Senturia & Frank, 2023). In contrast, transformer-based language models seem to have a more immediate bias for imposing linear relations (Huang et al., 2024; McCoy et al., 2020; but see also Ahuja et al., 2025).

Recent surveys indicate that OpenAI's o3-mini-high fails to reliably distinguish between syntax and semantics (Murphy, Leivada, et al., 2025), with other studies

offering conservative assessments of the prospects for LLMs to capture human-like linguistic intelligence (Dentella et al., 2024; Murphy, de Villiers, et al., 2025; for alternative perspectives, see Goldberg et al., 2025). Some of the neurobiological evidence cited by Parr and colleagues has been argued to be problematic through overlooked confounds (Hadidi et al., 2025).

Meanwhile, Slaats and Martin (2025) demonstrate why surprisal does not lead to natural language syntax, arguing that 'surprisal and entropy are not, by themselves, suitable as *explanans* for the core capacities of language;' probabilistic information is rather a 'cue to the next level of abstraction' (Slaats & Martin, 2025). We might consult here hierarchical generative syntax where probabilities sit on trees, not strings (Hunter, 2019).

Parr and colleagues could potentially enrich their research program by importing syntactic/parsing tree-generating operations into their latent state space, preserving the explanatory successes of generative grammar while retaining probabilistic machinery for prediction and memory. Deep temporal models can host typed variables which renders the transition to modeling certain features of syntactic recursion easier. It is true that a higher-level latent state can remain unchanged, while a lower-level sequence unfolds – this gives us temporal abstraction. What it does not yet buy us is *configurational simultaneity*, where multiple terminals are co-present in the same derivational moment even though they are pronounced at different times (necessary for agreement and binding). Perhaps, the higher-level sentence state could spawn an event frame within which lower-level factors are synchronized but unordered. If we also have an explicit linearization policy conditioned on the syntactic workspace state, a workspace tensor, an internal symbolic stack, or pointer variables, we might begin to approximate something like a push-down automata, with variational inference replacing deterministic stack operations. One trick here might be to make the latent state at each time-step

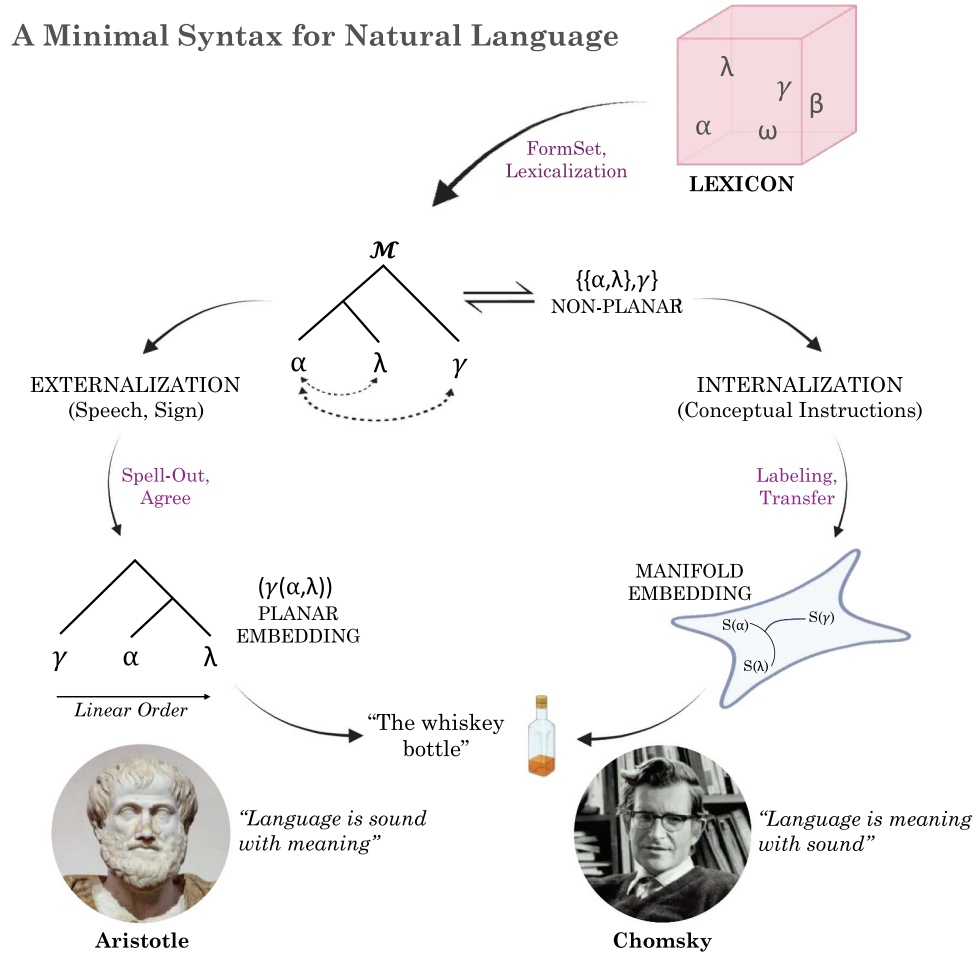


Figure 1. The architecture of the human language faculty (presented here at a higher altitude of generalization than specialist presentations in Chomsky et al., 2023; Marcolli et al., 2025; Murphy et al., 2024). This model is strictly atemporal; MERGE is a *free non-associative commutative magma*, and generates nonplanar trees. This algebraic formulation also provides potential (though not strict) constraints on what the neural code for syntax should look like. During internalization, structures are mapped to distinct conceptual systems (likely ‘core knowledge systems,’ Spelke & Kinzler, 2007) with their own distinct manifold embeddings. Lexical elements (α, λ, γ) are mapped to a workspace via MERGE ($\mathcal{M}$), which is then sent to interpretive systems and categorized (‘labeled’), and/or are ‘spelled-out’ at externalization systems for speech, gesture, etc. For example, ‘the whiskey bottle’ would be formed via $\{\textit{whiskey}, \textit{bottle}\}$ being merged with $\{\textit{the}\}$, to form a hierarchical object that is externalized in different orders depending on the language in question (i.e., not all languages adhere to the English ‘Determiner-Noun’ order, but all human speakers converge on the same coordinates in semantic space for the phrase). The quotations at the bottom contrast opposing views on language design (see Chomsky, 2011). Portrait photographs reproduced from creative commons (Wikimedia Commons; commons.wikimedia.org): public domain photo of marble bust of Aristotle (left). Public domain photo of Chomsky from Andrea Womack (c. 1973) (right).

a representation of ‘current derivational state,’ not just ‘what word came before’ – a left-corner minimalist grammar parser might then fall out naturally from the same inference machinery that already drives attention and memory.

While it is possible that human syntax might algorithmically be implemented by transformers (Marcolli et al., 2025), there may be a kind of competence-performance gap, or the format of representation may be entirely distinct. It is not immediately clear how we might begin to iron out these concerns – but the exciting approach from Parr and colleagues is nevertheless

a fruitful step in this direction. For example, there may be certain ways to reconcile predictive processing with the present observations about syntax (e.g., non-Markovian memory reconciled with syntactic derivational history stack; attention gain reconciled with derivation phrase head selection; hidden state reconciled with syntactic workspace).

With all this said, I should also acknowledge the possibility that the algebraic recipe for syntax provided by modern mathematical linguistics (Figure 1) may be insufficient for mapping the psycholinguistic and neural properties of language. The brain may not in fact

construct tree-structures in real time, and theoretical models of syntax may end up describing our knowledge of language at some higher level of generalized abstraction. As Captain Barbosa once described the pirates' code, it may have less in common with hard-coded rules and be more akin to a set of 'guidelines.'

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